

This lecture provides an introduction to Python programming, covering fundamental concepts such as variables, data types, and data structures. The lecture begins with a discussion of system setup and then transitions into the basics of Python programming. Key topics include:

- Variables:
 - Creating variables and assigning values (e.g., `a = 10`).
 - Understanding that variables are names that store data in memory.
 - Python is a self-typed language, meaning you don't need to explicitly declare data types.
- Data Types:
 - Integer: Whole numbers (e.g., `10`).
 - Float: Decimal numbers (e.g., `34.5657`).
 - String: Textual data enclosed in single, double, or triple quotes (e.g., `'sudh'`, `"sudhanshu"`, `'''multi-line string'''`).
 - Discusses the use of single, double, and triple quotes for strings, including handling single quotes within strings by using double quotes and vice versa.
 - Triple quotes are used for multi-line strings.
 - Boolean: `True` or `False` values (case-sensitive).
 - Internally, `True` is represented as `1` and `False` as `0`.
 - Complex: Numbers with a real and imaginary part (e.g., `6 + 8j`).
 - Uses `j` to represent the imaginary unit.
 - Demonstrates how to access the real and imaginary parts using `.real` and `.imag`.
- Data Structures:
 - List: An ordered, mutable collection of items enclosed in square brackets (e.g., `[10, "sudh", 34.5657, True]`).
 - Lists can hold heterogeneous data types.
 - Lists can be nested.
 - Tuple: An ordered, immutable collection of items enclosed in parentheses (e.g., `(10, "sudh", 34.5657, True)`).
 - Explains the difference between lists and tuples in terms of mutability.
 - Set: An unordered collection of unique items enclosed in curly braces (e.g., `{10, 20, 30}`).
 - Sets automatically remove duplicate values.
 - Python is case-sensitive, so `Variable`, `variable`, and `VARIABLE` are treated as different identifiers.
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